StockPulse — Product Requirements Document

Tagline: One dashboard for every store and warehouse. Your daily numbers, delivered on WhatsApp.

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1. Executive Summary

StockPulse is a WhatsApp-native inventory and price-intelligence tool built for small and medium business (SMB) owners in Pakistan who run multiple sales channels — online stores and a physical warehouse/shop — and currently have no single, reliable way to see their stock, track supplier price changes, or know what's happening in their business without manually checking each channel separately.

The product combines a simple multi-channel dashboard, AI-driven purchase-price anomaly detection, and two-way WhatsApp interaction, so an owner never has to open an app to know the state of their business — their business comes to them.

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2. The Problem

- SMB owners running 2-3 online stores plus a physical warehouse have **no single view of stock** — they check each channel separately, manually, every day
- **Price creep from suppliers goes unnoticed** until it has already cost real money — there is currently no system that tracks purchase-price changes item by item and flags them
- Owners are **time-poor and not software-first** — they don't want to learn a new dashboard or log into an app daily; they already live in WhatsApp for nearly everything else in their business
- Existing inventory software is either **too expensive** (global enterprise tools), **not built for Pakistan** (India-focused tools), or **not proactive** (reports exist, but nobody tells the owner something is wrong until they go looking for it)

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3. Ideal Customer Persona (ICP)

Name (persona): Ahmed Raza — owner, General Store & 2 online storefronts

- **Age:** 30-45
- **Business:** Runs a physical shop/warehouse plus 2-3 online storefronts
  (could be grocery, clothing, electronics — category varies, structure doesn't)
- **Team:** 1-3 staff helping with stock and sales, no dedicated IT or accounts person
- **Current process:** Tracks stock across a mix of memory, a notebook,

  maybe one spreadsheet — nothing connected, nothing automatic
- **Relationship with technology:** Comfortable with WhatsApp daily, not comfortable with unfamiliar business software or dashboards with a learning curve
- **Core pain:** Doesn't find out a supplier raised prices until doing monthly totals — by then the cost is already absorbed. Doesn't know exactly what's in the warehouse versus what's actually on the shop floor without physically checking or calling someone
- **What motivates a purchase decision:** Trust and simplicity over features. Will not adopt something that requires training or feels like "enterprise software." Price-sensitive — needs to see the cost clearly justified against what they're currently losing to unnoticed price changes and wasted time
- **Quote (persona voice):** *"I don't have time to check four different places to know what's going on. Just tell me on WhatsApp."*

Why this persona is accurate, not a guess
This persona is built directly from the actual gap identified against existing competitors (Section 8) — global tools (Zoho, Extensiv) are priced and designed for businesses with dedicated staff to run them; India-focused tools (Vyapar, Syncli) assume GST/UPI workflows that don't exist in Pakistan. The persona reflects the specific owner these tools currently fail: WhatsApp-fluent, time-poor, price-sensitive, and running a hybrid online+physical operation rather than a single clean channel. This will be validated directly with 5-10 real shop owners during the pilot (Section 11) before being treated as final.

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4. The Solution

One system that watches stock and prices across every store and the warehouse, and sends the owner what they need to know on WhatsApp — automatically, in their own language, without requiring them to open an app.

Why WhatsApp specifically
Because the target customer already lives there. Every competing tool requires the owner to change their behavior (log into a dashboard); StockPulse fits into behavior that already exists. This is the single biggest reason this approach is more accurate for this ICP than a traditional dashboard-first product.

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5. Core Features (Complete Detail)

5.1 Multi-Channel Dashboard
- Single view pulling stock and sales data from every connected online store plus the physical warehouse/shop
- Built for non-technical users: every screen leads with a plain-language status line (e.g. "Everything looks fine today") before any raw numbers — see Section 7 for full value-proposition rationale
- Key stats always visible: total stock value, total distinct/different

  items, active price alerts, low-stock count
- **Item breakdown view:** clicking into "different items" opens a full list showing, per product, the total quantity and its exact split between warehouse and shop floor (e.g. "Biscuits — 48 total: Warehouse 30, Shop 18") — so the owner always knows not just how much stock exists, but *where* it physically is
- **Low stock view:** clicking into "low stock" opens a filtered list of exactly which items have fallen below their restock threshold

5.2 WhatsApp Integration — Daily Report
- Automatically generated and sent once per day (evening, configurable)
- Contains: stock snapshot, biggest price change of the day, one actionable flag (e.g. "reorder chips — 3 days of stock left at current pace")
- Sent in the owner's chosen language (see 5.7)
- Delivered as a Utility-category WhatsApp message (proactive, business-initiated)

5.3 WhatsApp Integration — Two-Way Q&A
- Owner can message the number directly — e.g. "biscuits ka stock kitna hai?" or "is mahine sabse zyada profit kis product se hua?" — and receive an instant, accurate answer pulled from their real data
- This is what makes the product feel like "someone watching the shop," not just a report generator
- Classified as a Service-category message under WhatsApp's messaging rules
- **Voice note support:** WhatsApp already supports voice notes natively — the Q&A feature accepts a voice note the same way it accepts typed text, transcribes it, and replies normally. This directly serves staff who aren't comfortable typing, especially in a language other than their own

5.4 AI Price-Anomaly Detection
- The moment a new purchase price is entered for any product, it is automatically compared against the last several purchases of that same item
- If the price has moved beyond a set threshold (e.g. ±10%), an alert is triggered **immediately** — not surfaced only in a monthly report — catching supplier overcharging, data-entry mistakes, or genuine market price shifts while they're still actionable
- This is proactive by design: the owner is told, they don't have to go looking

5.5 Price Comparison — Two Precise, Independent Functions

(a) Instant, event-driven comparison — the core function:
- Trigger: a new purchase price is entered for any product
- Compares: that new price against the **same product's last recorded purchase price** — not a calendar date, the actual last purchase, whenever it happened
- Output: % change = (new − old) ÷ old × 100; if it exceeds the threshold (~10%), an immediate WhatsApp alert fires — "📈 Biscuits: PKR 145, up 12% from last purchase"
- This is not rigid or scheduled — it runs exactly when stock is added,

regardless of day of month

(b) Monthly summary chart — a separate, scheduled rollup:
- Generated automatically at the end of each calendar month
- Compares that month's purchase prices per product against the previous month's, using whatever purchases actually occurred during the month
- Sent as one chart image (bar comparison) via WhatsApp attachment
- Runs independently of (a) — this is a visual summary, not a replacement for the instant alerts

In the dashboard: a dedicated Price History screen per product, showing the same bar comparison plus explicit stat rows for current price and % change, available anytime (not only at month-end)

5.6 Data Backup (corrected architecture)
- **Live data must sit in a real persistent hosted database** — not a local SQLite file on Vercel's serverless hosting, which is not guaranteed to persist between deployments or requests. This was an identified gap: backing up data that itself sits on unstable storage does not provide real protection.
- Automatic daily backup: a full snapshot exported to a **separate storage location**, stamped with the exact date and time
- **Backup History screen** (in Settings): shows the complete list of past backups — date, time, status — not only the most recent one
- Backup status is also surfaced directly on the Dashboard — "Last backed up: [date/time] ✓" — so the owner has visible, ongoing proof, not just a promise
- Purpose: protect against data loss, corruption, or server issues — the owner should never be able to lose their business records

5.7 Language Selection
- Chosen once during first-time setup, changeable anytime afterward in Settings
- Full dropdown of supported languages: English, Urdu, Chinese, German, Arabic, and others — not a limited hardcoded set
- The selected language applies everywhere the owner interacts with the product: WhatsApp daily reports, price alerts, Q&A replies, and the dashboard interface itself
- Dropdown interaction: collapsed by default (e.g. "English ▾"), opens into a clean list with a smooth, soft hover highlight — this is a deliberate UX detail, not incidental

5.8 Multi-Tenant Architecture (Data Isolation)
- One shared core application ("skeleton") — same features, same code, same structure — used by every shop owner
- Every shop owner's data is completely private and isolated from every other owner's data. A grocery owner's biscuits/juices/chocolates and a clothing owner's shirts/sizes never mix, never become visible to each other, and are stored in entirely separate data spaces
- Each owner defines their own product categories inside their own private space — the system does not assume or hardcode a business type, so it works identically well for groceries, clothing, or any other retail category

- This is standard SaaS multi-tenancy: one product, many private users

5.9 Owner vs. Staff Access (revised — major access model decision)

Owners:
- Full access to the entire web dashboard — every screen, chart, and report
- Log in via WhatsApp OTP (one-time code sent via WhatsApp, no passwords) — see Section 10 for the security requirement this satisfies
- Daily reports, price-comparison charts, and PDFs are sent **only** to owner WhatsApp numbers, never to staff
- **Multiple owners are supported**, with a configurable access type per additional owner — "Equal owner" (full control, including adding/removing other owners) or "Co-owner" (full dashboard/report access, but cannot remove other owners or change critical settings). This is intentionally left configurable rather than fixed to one model, since the right structure may vary by business.

Staff (corrected — real in-app access, restricted by role):
- Staff log in the **same way as owners** — WhatsApp OTP, one login system for everyone, no passwords anywhere
- Staff **never see the main Dashboard/overview screen** (stock value, status line, charts, recent price changes) — that stays owner-only
- Staff **can** reach functional screens needed for their job — primarily Add Product, with voice input already available there for anyone not comfortable typing
- "Can add stock" → can open Add Product and submit; "View only" → can view a basic read-only stock list, cannot add or edit
- Staff never reach Price History, Settings, or Owner/Staff management regardless of permission level
- Staff **never** receive the daily report, price-comparison charts, or PDFs — owner-only
- Added by **name and WhatsApp number together** (name required)
- WhatsApp conversation remains available as an optional additional channel, but is no longer the only way staff can use the product

5.9B Voice/Microphone Input (Add Product screen)
- A microphone button next to the Add Product form fields, allowing product name, quantity, and other fields to be filled by speaking instead of typing
- Exists specifically to serve staff who aren't comfortable typing, or typing in a language they're less fluent in — the same motivation behind voice-note support in WhatsApp Q&A (5.3)

5.9C Stock Deduction & Automatic Total Recalculation
- Each location (Warehouse, Shop, Online Store) holds its own quantity for a product
- A sale or stock movement out of a specific location (e.g. 5 units sold from the Shop) must be recordable, decreasing that location's quantity
- The **total stock quantity recalculates automatically** as the sum across all locations — the owner never manually re-totals stock themselves
- In the pilot, this deduction is entered manually; a connected online

  store would eventually sync this automatically in a later phase

5.10 AI Integration — Where AI Is Actually Used
- **Price-anomaly detection** (5.4): AI evaluates each new purchase price against historical purchase data for that item and flags meaningful deviations
- **Natural-language WhatsApp Q&A** (5.3): AI interprets the owner's plain-language question (in their selected language) and returns an accurate answer sourced from their real, private data — not a generic/hallucinated response
- **Daily report generation** (5.2): AI drafts the plain-language summary and selects the single most relevant actionable flag for that day, rather than dumping every data point
- AI is used to reduce what the owner has to read and interpret, not to add complexity — every AI-generated output is short, direct, and actionable

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6. Screens (Confirmed Set)

Screen	Purpose
Setup	First-run: language, business name, getting-started checklist, sample-data trial option
Dashboard	Main screen — status line, hero stock value stat, different items / price alerts / low stock tiles, recent price changes
Item Breakdown	Full item list with warehouse vs. shop quantity split
Low Stock	Filtered list of items below restock threshold
Add Product	Product name, category (free text), location, quantity, purchase price, optional variant, optional photo (hidden behind a "want to add a photo?" button, never required), microphone/voice input for any field
Price History	Per-product chart, this month vs. last month, % change
Staff Access	Add/view staff by name AND WhatsApp number together, plus permission level
Settings	Language, business name, backup status

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7. Why Customers Want This (Value Proposition, Explicit)

1. **It removes a daily chore.** Owners currently check multiple channels manually — this replaces that with one automatic message
2. **It catches losses they can't currently see.** Price creep is invisible until a monthly total; this catches it the same day it happens
3. **It requires no new habit.** No new app to remember to open — it uses WhatsApp, which the owner already checks constantly
4. **It's built for their actual language and business context** — not a global or India-first tool translated after the fact
5. **It's affordable relative to what it replaces** — cheaper than

enterprise tools, while doing the one thing that actually matters to this owner (visibility + proactive alerts), rather than a long feature list they'll never use

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8. Competitive Landscape

Product	Strength	Why it falls short for this ICP
Zoho Inventory	Mature multi-channel dashboard	Enterprise-priced (PKR 5,600+/month), English-only, not WhatsApp-native
Extensiv / Brightpearl	Strong cross-channel reporting	Same — global/enterprise pricing and complexity
Vyapar	WhatsApp-integrated, invoicing	Built around India's GST/UPI ecosystem, not Pakistan's
Syncli	WhatsApp catalog sync	Same India-first limitation

The gap: no existing product combines WhatsApp-first two-way interaction, proactive AI price alerts, and Pakistan-specific setup (language, currency, payment methods) in one product. This is the specific, validated opening StockPulse is built for.

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9. WhatsApp BSP Strategy

- **Pilot phase:** Twilio — free sandbox testing, pay-as-you-go, no upfront cost, fits a developer-led build
- **Post-pilot, going live in Pakistan:** switch to WeTarseel — PKR-native billing, local support, ~PKR 14,000/month platform fee

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10. Non-Functional Requirements

- **Data isolation:** strict multi-tenant separation (Section 5.8) — no shop owner's data is ever visible to another
- **Reliability:** automatic daily backups (Section 5.6), no data loss even in the event of a crash or corruption
- **Accessibility for non-technical users:** every screen must be usable by someone with no prior business-software experience — plain-language status lines, no jargon, no unexplained icons
- **Accessibility for non-readers/non-typists (standing rule):** every future feature must be checked against "can someone who cannot read or cannot type actually use this?" Any text-entry field needs a microphone/voice alternative; any list a person needs to recognize products from needs photo thumbnails shown, not just stored. This applies going forward to every screen, not only the ones already identified.
- **Performance:** WhatsApp responses (Q&A) should feel near-instant; the dashboard should load quickly even on modest mobile connections, since owners may check it on their phone

- **Responsiveness:** fully usable on mobile, not just desktop
- **No external dashboards required for verification:** checking that the app is healthy (database connected, WhatsApp connected, backup status) must never require logging into Turso, Twilio, Vercel, or any other external service. A System Status panel inside Settings shows this in plain language with simple visual indicators — this applies equally to the builder's own daily checks and to anything a future non-technical owner might need to see.

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11. Demo / Pilot Plan

Goal: validate the two core hypotheses — that owners actually want proactive price alerts, and that WhatsApp-native interaction beats another login — before building the full product.

In scope for the pilot:
- Manual stock/price entry for one store/warehouse per owner
- Daily WhatsApp report
- Price-anomaly alert on new price entry
- Basic WhatsApp Q&A
- Simple database (not a large production system)
- Twilio free sandbox for WhatsApp testing (zero pilot cost)

Out of scope for the pilot (later phases):
- Real multi-channel API integrations (live store syncing)
- Voice-note stock entry
- Supplier comparison
- Shrinkage/mismatch detection

Who: 5-10 real shop owners, recruited directly, given hands-on access and asked for honest feedback — pricing may be offered free during this phase specifically in exchange for candid input

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12. Success Metrics (Pilot)

- % of pilot owners who check the WhatsApp report without being reminded
- Number of unprompted Q&A questions asked per owner per week (signals real engagement, not just passive reading)
- Number of genuine price anomalies caught that the owner confirms they would not have otherwise noticed
- Willingness-to-pay signal from direct conversation at the end of the pilot period, against the proposed pricing tiers

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13. Risks & Open Items

- **Pricing is not yet validated** with real owners — treated as a hypothesis until the pilot confirms it
- **WhatsApp Service message pricing change (Oct 1, 2026)** affects live

  cost structure — needs to be re-checked against actual BSP billing once past the pilot
- **Customer acquisition** — which 5-10 shop owners, and how they're approached, is not yet finalized
- **Scaling data isolation and backup infrastructure** beyond pilot scale has not yet been architected in detail — current plan is sufficient for 5-10 pilot users, will need revisiting before wider launch
- **Twilio Sandbox is development-only, not sellable as-is.** The current WhatsApp connection requires each number to manually "join" via a code — fine for the builder's own testing, but not acceptable to put in front of real paying customers. Must move to a real approved WhatsApp Business number via a BSP before selling to anyone outside testing.
- **No payment/trial gating exists yet.** Anyone with the live link can currently sign up and use the full real app for free, indefinitely. Acceptable during the free pilot, but this is the critical blocker before the product can actually be sold.
- **OTP rate-limiting was a missing safeguard**, now added in Prompt 8 — without it, the login system could be abused to spam a number with codes or run up Twilio costs.

This document reflects every decision made through the current planning stage. It should be treated as the source of truth for scope — any new decision should be added here, not left undocumented.